Muhammet Celal Gokalp

Mechanical, Systems & Test Engineer | Rotating Machinery, Controls & Test-Bed Validation

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PROFESSIONAL SUMMARY

Mechanical engineer (BSc Mechanical Engineering; MSc Sustainable Automotive Engineering, WMG Warwick) with 8 years in systems engineering, control validation, test-bed / vehicle testing and fault-finding at AVL Powertrain UK and FiveAI / Bosch. My background includes stator thermal modelling and thermocouple test design for rotating electrical machines (MotorCAD, Portunus, Simulink), hands-on electrical checks on returned range-extender generators at the LEVC factory, PID controller calibration on AVL's DrivingCube dyno test bed, and requirements/safety engineering (DOORS, ASPICE V-model, ISO 26262, FMEA). I have not yet worked on gas, steam or wind turbines, and I am keen to move from automotive into the turbine and energy sector. I love hands-on work with hardware and test beds, learn new systems quickly, and am immediately available in Coventry.

WORK EXPERIENCE

Lead Application Engineer (Solutions & Integration)

10/2021 – 11/2026

FiveAI / Bosch — Cambridge, United Kingdom

- **Leading a team & work packages:** Led a 5-person engineering team for a year (technical lead): selected team members, split work packages by tool experience, held weekly progress meetings, and reviewed deliveries against specifications.
- **Performance KPIs & root cause analysis:** Defined pass/fail and performance KPI criteria across 1,000+ hours of test data, wrote 120+ parameterised test scenarios, and diagnosed errors from logs between APIs and simulation tools across partner teams (Volkswagen Group ADA project).
- **Python tools & data pipelines:** Built Python tools with AI help (JSON parameter GUI, uploader, data pipelines into dashboards) and worked in Linux in the project monorepo with a dev container and CI test pipeline.

ADAS Systems Engineer (Feature Owner)

11/2018 – 10/2021

AVL Powertrain UK — Coventry, United Kingdom

- **Generator & rotating-machine fault-finding (LEVC factory):** Inspected ~20 returned faulty range-extender generators at the London Taxi factory following a test procedure: performed multimeter checks at specified connector pins (phase-to-phase resistance, continuity), powered the control unit from a low-voltage bench supply to download fault logs with the supplier's logging tool, and sent measurements and data to the supplier for root cause analysis.
- **Dyno test bed & PID calibration (AVL DrivingCube):** Compared a real system, a chassis dyno with sensor emulation and a MATLAB / Simulink + IPG CarMaker model. Fitted external ground-truth instrumentation (GNSS), captured test data at HORIBA MIRA and on the airstrip, calibrated the model and swept PID controller gains, and monitored logging health during test-bed runs.
- **Requirements, state machines & safety:** Based on-site at Jaguar Land Rover for 2 years as feature owner for three safety-related features. Wrote use cases, system state diagrams, module input/output signal lists and high- to low-level requirements in DOORS and MKS Integrity (ASPICE V-model). Wrote ISO 26262 safety requirements and took part in an FMEA.
- **Verification & supplier root cause:** Wrote test cases for each system state and ran verification in vehicles and on the proving ground, using Vector CANoe and CANape and tracking defects in RTC / RQM. Synchronised camera, bus and sensor logs to trace a boundary timing fault and verified the supplier's calibration fix on site. Diagnosed module faults via fault codes, fuse checks and a 12 V bench supply.

Mechanical Engineering Intern, TEMSA Global, 2014: calculated power and torque requirements for a new electric bus powertrain in MATLAB / Simulink, and supported quality work on the production line.

EDUCATION

MSc Sustainable Automotive Engineering, WMG, University of Warwick, 2017 – 2018 (2:1 Merit). Dissertation: designed a test to find the unknown values of a thermal model of an electric motor stator (type-K thermocouple placement, enclosure, method) using MotorCAD's thermal network, Portunus and MATLAB / Simulink.

BSc Mechanical Engineering, Ege University, Izmir, 2011 – 2017. Dissertation: team project to design and build a 6-DOF Stewart motion platform (SolidWorks, MATLAB state-space model, Arduino); wired the 6 servo actuators, wrote Arduino code to command set angles, and plotted encoder feedback signals.

SKILLS & HANDS-ON EXPERIENCE

Rotating machines & thermal	Electric motor stator thermal network modelling & thermocouple test design (MotorCAD, Portunus, Simulink – MSc); range-extender generator resistance, continuity & log checks (LEVC); powertrain sizing (TEMSA)
Testing, test beds & controls	AVL DrivingCube (ViL dyno test bed), vehicle & proving ground testing, PID gain sweeps, MATLAB / Simulink, Vector CANoe / CANape, multimeter & 12 V bench supply fault-finding (all at AVL); KPI & log analysis (AVL & FiveAI)
Systems & safety	Requirements decomposition, state diagrams, module signal lists, DOORS, MKS Integrity, RTC / RQM, ASPICE V-model, FMEA & ISO 26262 (all at AVL); Jira, Confluence, technical lead for 5 engineers (FiveAI)
Software & hands-on	Python (tools and pipelines built with AI help), Linux (5 years), Git, SolidWorks. Hands-on: repairs own cars (multimeter fuse probing, torque wrench, calipers, micrometers), soldered motor cables to a PCB, builds PCs
Languages & details	English (C2), Turkish (native). British citizen, full UK driving licence, based in Coventry (close to Warwick / Rugby / Derby), immediately available